Examiner only

2. A teacher was marking the exercise book of a student who had completed a topic of work on acids and alkalis.

(a) The table below shows the results the student had recorded from an experiment to investigate the pH values of some common laboratory chemicals.

The teacher has circled three errors for the student to correct.

Chemical name	Chemical formula	Colour with universal indicator	pH	Acid, alkali or neutral
sulfuric acid	H_2SO_4	green	1	acid
ethanoic acid	CH_3COOH	orange	4	alkali **error 1**
calcium hydroxide	$Ca(OH)_2$	purple	12	alkali
water	H_2O	green	5 **error 2**	neutral
sodium carbonate	$NaCO_3$ **error 3**	blue	10	alkali

(i) Correct the errors the teacher has circled. [3]

Correction to **error 1** ....

Correction to **error 2** ....

Correction to **error 3** ....

(ii) There is another error in the table that the teacher has not spotted.
Circle this error **in the table**. [1]

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(b) Another piece of work in the exercise book showed the results of some tests that the student had carried out to identify some of the ions in the chemicals being investigated. However the table had not been fully completed by the student.

Ion being identified	Test	Result
carbonate CO_3^{2-}	add hydrochloric acid	bubbles formed as .... gas is produced
sulfate SO_4^{2-}	add .... solution	white precipitate is formed

(i) Use the names of chemicals from the box below to complete the table. [2]

silver nitrate	**hydrogen**	**sodium hydroxide**
barium chloride	**oxygen**	
universal indicator	**carbon dioxide**	**chlorine**

(ii) Describe the test that can be used to identify the gas produced when hydrochloric acid is added to a compound containing carbonate ions. [1]

....

....

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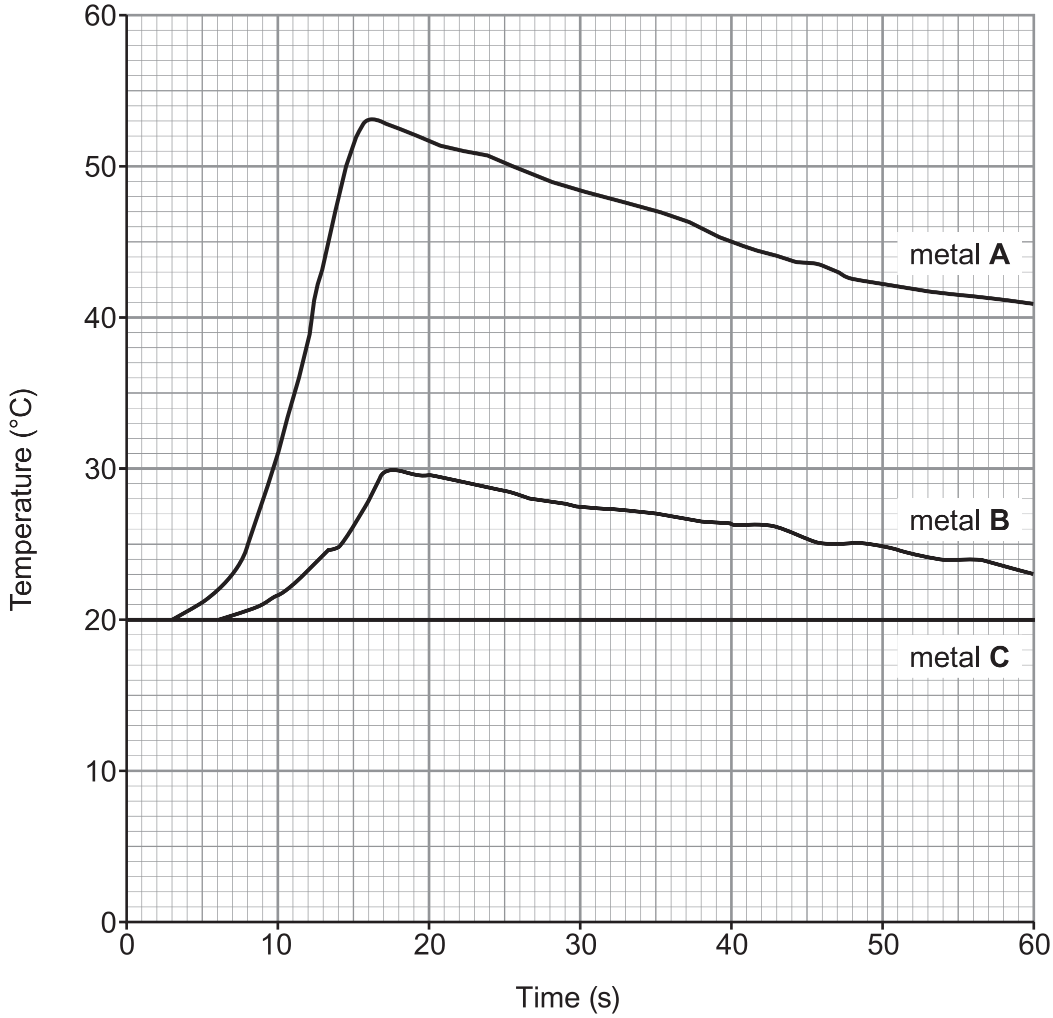
(c) The final piece of work marked by the teacher was an experiment that had been completed to investigate the temperature change when different metals react with hydrochloric acid.

The graphs of the results obtained by the student are shown below.



(i) Use the graphs to give

I. the **letter** of the metal that did not react with hydrochloric acid. [1]

....

II. the maximum temperature **rise** for metal **A**. [1]

.... °C

(ii) State the term used to describe a reaction that gives out heat. [1]

....

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(d) When metals react with acids they form a salt and hydrogen gas.

(i) Complete the word equation for the reaction between magnesium and sulfuric acid. [1]

magnesium + sulfuric acid ⟶ .... + hydrogen

(ii) Give the test that can be used to identify hydrogen gas. Include the expected observation. [1]

....

....

(iii) The salt formed when zinc reacts with hydrochloric acid is zinc chloride.

I. Give the formulae of the ions present in zinc chloride. [1]

.... and ....

II. Give the formula of zinc chloride. [1]

....

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